5 November 2026

To the Editors, Journal of Process Control

Dear Editors,

I am submitting the manuscript "A credibility-weighted hybrid soft sensor and uncertainty-consuming robust real-time optimization for silica scaling control in Philippine geothermal fields: design, negative results, and out-of-sample validation" for consideration as a full-length research paper.

The paper sits squarely in this journal's scope: soft sensing under sparse labels, economic real-time optimization under plant-model mismatch, and the interaction of the two in closed loop. The modifier-adaptation lineage it builds on and tests against is work published by this journal's own community. The application, silica scaling control in Philippine liquid-dominated geothermal fields, supplies a constraint process of unusual severity: a fourth-order peak-rate law, an induction delay, a non-monotonic inhibitor actuator, and a temperature lever that multiplies the scaling rate by a factor of 101 across the decision range.

The contribution is the coupling that current practice lacks. A hybrid soft sensor weights its learned residual by leave-one-out credibility, so the correction fades to the physics prior when data has not earned it, and the same computation exports the residual's uncertainty. A vertex-robust economic optimizer then consumes both, the uncertainty as a multiplicative constraint back-off. The mismatch this addresses is measured rather than assumed: the transcribed classical kinetic law mispredicts an independent 2024 laboratory corpus by a factor of six at the median, and that measured gap calibrates the adverse stress test. Across two regimes, benign random mismatch and the sixfold stress, the proposed loop is the only tested policy essentially free of constraint violations (zero violation days benign at +33% power uplift; 0.50 post-warm-up days under stress against 17.2 for the robust open loop). Two pre-registered negative results are reported as results: zero-shot residual transfer actively harms, and scalar modifier adaptation cannot stabilize a multiplicative error on a quartic constraint. Every physics component and the sensing pattern itself are validated out-of-sample, yielding two findings of independent geochemical interest and a design principle: identifiability precedes correction.

Every quantitative claim regenerates from a single gate script (94 automated checks across 13 modules, deterministic seeds throughout) against checksum-validated transcriptions of the primary sources; the repository accompanies the paper and an archival release is planned at acceptance.

The manuscript is approximately 5,948 words with six figures and three tables. It is my original work, is not under consideration at any other journal, and I am the sole author. I have no competing interests to declare. Reviewers with expertise in soft sensing and hybrid modeling, real-time optimization and modifier adaptation, or geothermal process chemistry would be well placed to assess it.

Thank you for considering this work.

Sincerely,
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